

Two obstructions to a uniform weighted stable-limit extension

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Abstract

Berkes and Tichy ask whether their uniform weighted subsequence theorem remains valid, with centering, when the limit random measure merely belongs to the attraction domain of a fixed stable law. Two different broad readings are false. First, if the fixed stable law may be nonsymmetric, alternating blocks of signs make the skewness oscillate even when the variables themselves are strictly stable. Second, under the broader symmetric regularly-varying-tail reading, the condition

$$\max_{k \leq N} |a_k| = o \left(\sum_{k \leq N} |a_k|^\alpha \right)^{1/\alpha}$$

does not control the true tail scale. For a symmetric law with $\mathbb{P}(|X| > x) = 1/(x \log x)$, we construct positive coefficient blocks satisfying the displayed condition such that infinitely many endpoint sums, divided by their only possible nondegenerate scale, converge to the original nonstable law X . Consequently no normalization and no centering can yield a stable limit. Both examples are iid, so passing to a subsequence cannot help.

1 Source question and exact scope

The source is I. Berkes and R. Tichy, *Lacunary series and stable distributions*, arXiv:1707.08890 [1]. Its Theorem 1.2 produces one subsequence that works for every real coefficient sequence satisfying the uniform asymptotic negligibility condition. Immediately afterward the authors ask whether the conclusion persists, with a suitable centering factor, when the limit random measure lies in the domain of normal attraction of a fixed stable distribution:

Theorem 1.2 *Let $0 < \alpha < 2$, $c > 0$ and let (X_n) be a determining sequence of r.v.'s with limit random measure μ . Assume that $\mu \in S(\alpha, c)$ with probability 1. Then there exists a subsequence (X_{n_k}) such that for any real sequence (a_k) satisfying*

$$\max_{1 \leq k \leq N} |a_k| = o(A_N), \quad A_N = \left(\sum_{k=1}^N |a_k|^\alpha \right)^{1/\alpha} \quad (1.10)$$

we have

$$A_N^{-1} \sum_{k=1}^N a_k X_{n_k} \xrightarrow{d} G_{\alpha, c}.$$

Condition (1.9) holds provided the corresponding (symmetric) distribution function F satisfies

$$1 - F(x) = c_1 x^{-\alpha} + \beta(x) x^{-\alpha}, \quad x > 0$$

where $c_1 > 0$ is a suitable constant, $\beta(x)$ is non-increasing for $x \geq x_0$ and $\lim_{x \rightarrow \infty} \beta(x) = 0$. (See Berkes and Dehling [4], Lemma 3.2.) Apart from the monotonicity condition, this is equivalent to the fact that F is in the domain of normal attraction of a symmetric stable distribution. (See e.g. Feller [14], p. 581.) It is natural to ask if the conclusion of Theorem 1.2 remains valid (with a suitable centering factor) assuming only that $\mu \in S$ a.s. where S denotes the domain of normal attraction of a fixed stable distribution. From the theory in [1] it follows that the answer is affirmative in the unweighted case $a_k = 1$, but in the uniform weighted case the question remains open. Symmetry plays no essential role in the proof of Theorem 1.2; it is used only in Lemma 2.2 and at the cost of minor changes in the proof, (1.9) can be replaced by a condition covering nonsymmetric distributions as well. But since we do not know the optimal condition, we restricted our investigations to the case (1.9) where the technical details are the simplest and the idea of the proof becomes the most transparent.

Given a sequence (X_n^*) of r.v.'s and a random measure μ defined on a probability space $(\Omega, \mathcal{F}, P)$ such that X_n^* are conditionally i.i.d. given μ with conditional distribution μ , the limit random measure of (Y^*) is easily seen to be μ . The sequence

There is a terminology boundary that matters. In strict usage, a normal domain has a pure $N^{1/\alpha}$ scale (up to a constant), whereas a general stable attraction domain permits a slowly varying factor. The source's uncompiled text after `\end{document}` formulates the intended extension using balanced regularly varying tails $x^{-\alpha}L(x)$. We therefore give two logically independent results:

1. a counterexample inside a strict normal domain when the fixed stable law is nonsymmetric; and
2. a symmetric counterexample to the broader regularly-varying-tail extension, even if the normalization and centering are chosen freely.

We do not claim that the second example refutes the strict symmetric normal- domain version.

2 Proof intuition

The source condition measures coefficients only through their raw ℓ^α energy. Two extra data become indispensable outside the symmetric pure-scale setting.

For a nonsymmetric stable law, a positive coefficient contributes one sign of the stable skewness and a negative coefficient contributes the opposite sign. Raw absolute energy forgets this information, so very long alternating sign blocks can force two different endpoint laws.

For a tail $x^{-\alpha}L(x)$ with nonconstant L , raw energy also forgets the scale L . When $L(x) \rightarrow 0$, many equal small coefficients can carry enormous raw energy while their random sum remains much smaller. We exploit that gap to insert a new coefficient which is negligible in raw energy but dominates the actual random sum. At each stage endpoint the weighted sum therefore looks like one copy of the original, deliberately nonstable, variable.

3 A convergence-of-types fact

We use the following standard elementary consequence of tightness.

Lemma 1 (Convergence of types). *Suppose $Y_j \xrightarrow{d} Y$, where Y is nondegenerate. If $r_j > 0$, $s_j \in \mathbb{R}$, and $r_j Y_j + s_j \xrightarrow{d} Z$ for a nondegenerate Z , then every subsequence has a further subsequence along which*

$$r_j \rightarrow r \in (0, \infty), \quad s_j \rightarrow s \in \mathbb{R}, \quad Z \stackrel{d}{=} rY + s.$$

In particular, if the transformed sequence has a nondegenerate limit, that limit is of the same type as Y .

Proof. Choose two continuity points of the law of Y whose distribution values are strictly between zero and one and distinct. Convergence of the corresponding quantiles prevents r_j from tending to zero or infinity whenever $r_j Y_j + s_j$ is tight and nondegenerate. Tightness then also bounds s_j . Extracting a convergent subsequence of (r_j, s_j) and applying Slutsky's theorem gives $Z \stackrel{d}{=} rY + s$. The same argument applies after starting from any subsequence. $\square$

4 The sign obstruction in a strict normal domain

Theorem 2 (Nonsymmetric counterexample). *There is an iid determining sequence whose constant limit random measure is a strictly 1/2-stable law, and a real coefficient sequence satisfying the source negligibility condition, for which no positive normalization and centering can produce one nondegenerate distributional limit. In particular, the uniform weighted conclusion is false if the fixed stable law in the question is allowed to be nonsymmetric.*

Proof. Let $X_1, X_2, \dots$ be iid copies of a positive strictly 1/2-stable random variable X . Thus

$$X_1 + \dots + X_m \stackrel{d}{=} m^2 X.$$

Choose signs $a_k \in \{-1, 1\}$ in alternating blocks. Choose the length of block j greater than j times the total length of all preceding blocks. If N_j is its right endpoint and the sign of that block is ε_j , then the proportion of indices $k \leq N_j$ with sign ε_j tends to one. The signs ε_j alternate.

For every N ,

$$A_N = \left(\sum_{k=1}^N |a_k|^{1/2} \right)^2 = N^2, \quad \frac{\max_{k \leq N} |a_k|}{A_N} = N^{-2} \rightarrow 0.$$

Let p_N and q_N be the proportions of positive and negative signs among $a_1, \dots, a_N$. Stability and independence give the exact identity

$$N^{-2} \sum_{k=1}^N a_k X_k \stackrel{d}{=} p_N^2 X^+ - q_N^2 X^-,$$

where X^+, X^- are independent copies of X . Hence at positive block endpoints the normalized sums converge to X , while at negative block endpoints they converge to $-X$.

Suppose arbitrary centerings and positive normalizers produced one nondegenerate limit along all N . Lemma 1, applied first to the positive and then to the negative endpoints, would make that limit both a positive affine transform of X and a positive affine transform of $-X$. The former is bounded below and unbounded above; the latter is bounded above and unbounded below. They cannot have the same nondegenerate law. Finally, an iid sequence is determining with constant limit random measure $\mathcal{L}(X)$, and every subsequence remains iid, so subsequence selection cannot remove the obstruction. $\square$

5 The slowly varying scale obstruction

We now retain symmetry and use only positive coefficients.

Theorem 3 (Balanced regular-variation counterexample). *There exist a symmetric law F in the general domain of attraction of a symmetric 1-stable law, an iid determining sequence (X_n) with law F , and positive coefficients (a_k) such that*

$$\frac{\max_{k \leq N} a_k}{\sum_{k=1}^N a_k} \longrightarrow 0,$$

but for no positive normalizers d_N and centerings c_N does

$$d_N^{-1} \left(\sum_{k=1}^N a_k X_k - c_N \right)$$

converge to a nondegenerate stable law. The same assertion holds after replacing (X_n) by any subsequence.

Proof. Let $R \geq e$ have survival function

$$\mathbb{P}(R > x) = \frac{1}{x \log x} \quad (x \geq e),$$

with the remaining probability placed at e , and let $X = \epsilon R$, where ϵ is an independent symmetric sign. Then

$$\mathbb{P}(|X| > x) = \frac{1}{x \log x} \quad (x \geq e).$$

The tails are balanced and regularly varying with exponent -1 . The classical stable domain-of-attraction theorem therefore puts X in the domain of attraction of a symmetric 1-stable law. More explicitly, if $b_m > e$ is defined by

$$b_m \log b_m = m,$$

then $b_m \sim m / \log m$ and

$$b_m^{-1}(X_1 + \dots + X_m)$$

converges in distribution to a symmetric 1-stable law; in particular this sequence is tight. The law of X itself is not stable: its tail is $x^{-1}/\log x$, whereas a non-Gaussian stable law has a nonzero constant multiple of a power tail (and a Gaussian stable law has a light tail).

We construct the coefficients recursively. Put $w_0 = 1$. At stage j , append m_j copies of w_{j-1} and then append one coefficient

$$w_j = \frac{w_{j-1}m_j}{\sqrt{\log m_j}}. \quad (1)$$

Let T_{j-1} denote the weighted sum through the preceding stage. When m_j is selected, T_{j-1} is already fixed in law, while the right side of (1) tends to infinity with m_j . We may therefore choose integers $m_j \rightarrow \infty$ so fast that

$$\mathbb{P}\{|T_{j-1}| > w_j/j\} < 1/j, \quad (\log m_j)^{-1/2} < 1/j. \quad (2)$$

Let $M_N = \max_{k \leq N} a_k$ and $A_N = \sum_{k \leq N} a_k$. During the block of m_j copies, M_N/A_N can only decrease from its value at the preceding endpoint. Upon appending w_j , it is at most

$$\frac{w_j}{m_j w_{j-1}} = \frac{1}{\sqrt{\log m_j}}.$$

Induction over the stages and (2) show $M_N/A_N \rightarrow 0$ over every index N .

At the endpoint of stage j , write

$$T_j = T_{j-1} + w_{j-1} \sum_{r=1}^{m_j} X_{j,r} + w_j X_j^*,$$

using fresh iid variables. The first term divided by w_j tends to zero in probability by (2). For the middle term, tightness at scale b_{m_j} and (1) give

$$\frac{w_{j-1}b_{m_j}}{w_j} \sim \frac{1}{\sqrt{\log m_j}} \rightarrow 0.$$

Consequently Slutsky's theorem yields

$$\frac{T_j}{w_j} \xrightarrow{d} X. \quad (3)$$

If some centerings c_N and positive normalizers d_N gave a nondegenerate stable limit along the full sequence, they would do so along the stage endpoints. Apply Lemma 1 to (3), with $r_j = w_j/d_{N_j}$ and $s_j = -c_{N_j}/d_{N_j}$. The alleged stable limit would have to be an affine transform of X , contradicting that X is not stable. As before, every subsequence of an iid sequence has the same joint laws, so the single deterministic coefficient construction defeats every selected subsequence. $\square$

6 Consequences, limitations, and novelty audit

Theorem 2 shows that “symmetry plays no essential role” needs a qualification when arbitrary real coefficients are allowed: one must at least control the signed α -energy, or allow the target skewness to depend on its limiting balance. Theorem 3 shows that balanced regular variation alone is also insufficient under the source coefficient condition. A viable general-domain theorem needs a negligibility condition in the actual tail scale, involving L as well as $|a_k|^\alpha$.

The slow-variation example is not in a strict symmetric normal domain because $L(x) = 1/\log x$ does not tend to a positive constant. Accordingly we do not claim to settle a version that simultaneously insists on a symmetric fixed stable law and strict normal-domain terminology. In that regime the leading small-characteristic-function exponent is aligned with the hypothesis used in the source proof, after the appropriate centering.

A bounded search on August 13, 2026 checked all four lightweight run indexes for the arXiv id, exact title, and the weighted stable-domain keywords. External exact-title and weighted-sum searches found the source and classical triangular-array results whose hypotheses explicitly retain the slowly varying characteristic factor, but no later paper resolving this source question. The construction here is explicit and self-contained, but priority remains subject to specialist review.

The packet is recommended for expert review as a likely valid pair of full scoped counterexamples. The principal audit points are the standard positive-stable scaling in Theorem 2, the recursive quantifiers in Theorem 3, and the distinction between strict normal attraction and general regularly varying attraction.

References

- [1] I. Berkes and R. Tichy, *Lacunary series and stable distributions*, Periodica Mathematica Hungarica 78 (2019), 47–61; arXiv:1707.08890.
- [2] W. Feller, *An Introduction to Probability Theory and Its Applications, Vol. II*, 2nd ed., Wiley, 1971, Chapter XVII.